DESIGNING A CANCER MEDICINE — ON A COMPUTER

Disarming the one faulty protein behind a rare cancer

A rare cancer called **EMC** is driven by a single broken protein, **NR4A3**. This work is a step-by-step plan — worked out entirely in computer simulations — for a medicine that **finds that protein and throws it away**, while leaving everything healthy untouched.

1

THE PROBLEM

One protein is stuck "on"

Cells are run by tiny machines called **proteins**. In this cancer, one protein (NR4A3) is jammed in the **on** position, like a light switch taped down — and it keeps telling the cell to grow into a tumour.

2

WHY IT'S HARD

There's no keyhole for a drug

Most drugs work like a **key in a lock** — they plug into a pocket on the protein. NR4A3's pocket is **collapsed shut**, so there's nowhere to put a key. For decades this made it "undruggable."

3

THE BIG IDEA

Don't block it — bin it

So we don't try to jam the machine. Instead we design a molecule that **grabs the bad protein and hands it to the cell's own garbage disposal**. To do that you only need a **handhold** — not a keyhole. This kind of drug is called a **degrader**.

4

THE BREAKTHROUGH

We caught the door fluttering open

Using physics simulations on powerful computers, we found the "shut" pocket isn't really frozen — it **breathes open for a moment about 1 time in 4**. That brief opening is exactly the handhold our degrader needs.

5

MAKING IT SAFE

Hit the bad one — spare its near-identical twins

NR4A3 has two **almost-identical sibling proteins** that the body genuinely needs (harming them can cause leukaemia or Parkinson's-like damage). The danger is a drug that can't tell them apart. We mapped a handful of **tiny bumps unique to the bad protein**, so the degrader can grab **only NR4A3** and leave the healthy twins alone. We even used AI to invent brand-new candidate molecules built to fit those bumps.

Where this stands — the honest version

Everything here was done **on computers** — no medicine has been made or tested in a lab yet. Think of it as a careful, evidence-backed **blueprint and feasibility case**: it shows the idea is sound and worth the next step. The goal of the paper is to **convince the wider research world this is worth taking into the lab.**

EMC = extraskeletal myxoid chondrosarcoma, a very rare cancer · **NR4A3** = the protein that drives it · **Degrader** = a drug that throws a protein away instead of blocking it